Complete polynomials with sparse squares: an improvement of Verdenius' exponent via overlapping towers

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Abstract

For a polynomial f let $\|f\|$ be its number of nonzero coefficients, and let $\hat{Q}(k) = \min\{\|f^2\| : f \in \mathbb{R}[x],\ \deg f = k-1,\ \|f\| = k\}$ (complete polynomials). Verdenius (1949) proved $\hat{Q}(k) \ll k^{\log 8/\log 13} = k^{0.81071\ldots}$ along a sequence of k, which appears to be the best known upper bound. We show $\hat{Q}(k) \le \frac{21}{9}\, k^{\log 9/\log 17}$, $\log 9/\log 17 = 0.77552\ldots$, for the infinite sequence $k = (20\cdot 17^K - 4)/16$. The construction is a tower $\prod_j p(x^{m^j})$ with *overlapping* base $m < \deg p + 1$, a real seed p of degree 20 whose square has a three-block support, certified by interval arithmetic, and a short ℓ-adic argument proving that the tower stays complete. The result concerns real coefficients only; for rational/integer coefficients we obtain no improvement over $\log 6/\log 9$.

1 Background

Rényi and Erdős asked how sparse f^2 can be when f has many terms. Erdős [3] proved $Q(k) < k^{1-c}$; Verdenius [9] obtained the exponent $\log 8/\log 13$ for real (complete) polynomials and $\log 6/\log 9$ over $\mathbb{Q}$. The lower bound $Q(k) \gg \log k$ is due to Schinzel and Zannier [7, 8], improving Schinzel. Coppersmith–Davenport [2] and Abbott [1] studied the smallest complete examples (degree 12). We are not aware of any improvement of Verdenius' upper bound; cf. [8, 5, 4].

2 The construction

Let $p(x) = \sum_{i=0}^{n} a_i x^i$ have all $a_i \neq 0$, let $S = \mathrm{supp}(p^2) \subset [0, 2n]$, and let $m \le n+1$. Put

$$P_K(x) = \prod_{j=0}^{K-1} p\big(x^{m^j}\big), \qquad \deg P_K = D_K = n\,\frac{m^K - 1}{m-1}.$$

Since $P_K^2 = \prod_j p^2(x^{m^j})$, we have $\mathrm{supp}(P_K^2) \subseteq \{\sum_{j<K} s_j m^j : s_j \in S\}$. With $m = n+1$ this is Verdenius' setting (P_K automatically complete). We take $m < n+1$, which creates more collisions in the digit sums but makes completeness non-trivial.

The seed. $n = 20$, $m = 17$, and p palindromic ($a_{20-i} = a_i$, $a_0 = 1$) with

$$\mathrm{supp}(p^2) = S = [0,6] \cup [17,23] \cup [34,40] = A + 17E, \qquad A = [0,6],\ E = \{0,1,2\}.$$

*Computations, certification and first draft produced with an AI assistant (Claude, Anthropic) under the author's direction; see §6. Code and data: `https://github.com/RobertoVetere/sparse-squares`.

By palindromicity it suffices to impose $[x^j]p^2 = 0$ for $j = 7, \dots, 16$: ten quadratic equations in $a_1, \dots, a_{10}$. (In general, $p \equiv \sqrt{L} \bmod x^{17}$ with $L = \sum_{j \le 6}[x^j]p^2$, and the equations express palindromicity of the truncated square-root series.) Total-degree homotopy gives 259 isolated complex solutions, 42 of them real with all $a_i \neq 0$. One of them (to 20 digits) is

$$a_0..a_{10} =$$
$$1,\ 1.0048830402228113379,\ 1.2538227425985270778,\ 0.46211051619273127379,\ 1.0221701367519609109,$$
$$0.37735544346794049561,\ 0.43548160112730375387,\ -1.3831004818966026266,\ 0.14704166870239142598,$$
$$0.99944106767275121901,\ -1.0658751424342707659, \quad a_{20-i} = a_i.$$

Lemma 1 (certificate)**.** *The system above has exactly one solution in the box of radius 10^{-60} around the 110-digit refinement of these values; it is real, all $a_i \neq 0$, and $\operatorname{supp}(p^2) = S$ exactly.*

Proof. Krawczyk's test in interval arithmetic (110 digits, `mpmath.iv`): $K(X) \subset \operatorname{int} X$. The intervals of all a_i and of $[x^j]p^2$, $j \in S$, exclude 0; zeros off S hold exactly by construction and palindromicity. Script: `certify/krawczyk.py`. □

Lemma 2 (support)**.** $\|P_K^2\| \le 7 \cdot 9^{K-1} \cdot 3 = 21 \cdot 9^{K-1}$.

Proof. Write $s_j = \alpha_j + 17\varepsilon_j$, $\alpha_j \in [0, 6]$, $\varepsilon_j \in \{0, 1, 2\}$. Then $\sum_j s_j 17^j = \alpha_0 + \sum_{j=1}^{K-1}(\alpha_j + \varepsilon_{j-1})17^j + \varepsilon_{K-1}17^K$ and $\alpha_j + \varepsilon_{j-1} \in [0, 8]$. The number of such values is at most $7 \cdot 9^{K-1} \cdot 3$. □

Lemma 3 (unique minimal digit sum)**.** *Let $m - 1 \le n < 2m$. Every $N \in [0, D_K]$ has a representation $N = \sum_{j<K} i_j m^j$ with $0 \le i_j \le n$, and the one minimising $\sum_j i_j$ is unique.*

Proof. Existence is clear since $n \ge m - 1$. Let $r = (i_j) \neq r' = (i'_j)$ both be minimal and let j be the least index where they differ. Then $i_j \equiv i'_j \pmod m$ and $|i_j - i'_j| < 2m$, so w.l.o.g. $i_j = i'_j + m$. If some $l > j$ has $i_l < n$, take the least such t and replace $i_j \mapsto i_j - m$, $i_l \mapsto i_l + 1 - m$ ($j < l < t$), $i_t \mapsto i_t + 1$. All new digits lie in $[0, n]$, the value is unchanged, and the digit sum drops by $(t - j)(m - 1) > 0$, which contradicts minimality of r. Otherwise $i_l = n$ for all $l > j$, and then $\sum_{l>j} i'_l m^l = m^{j+1} + n\sum_{j<l<K} m^l$, which exceeds the largest value representable by digits $\le n$ in positions $j + 1, \dots, K - 1$. Contradiction. □

Lemma 4 (completeness)**.** *Let $F = \mathbb{Q}(a_0, \dots, a_n) \subset \mathbb{R}$. Choose a prime $\mathfrak{p}$ of F with $v_\mathfrak{p}(a_i) = 0$ for all i (all but finitely many primes qualify), and let ℓ be the rational prime below $\mathfrak{p}$. Then for $p_\ell(x) = p(\ell x)$, every coefficient of $P_K = \prod_{j<K} p_\ell(x^{m^j})$ in degrees $0, \dots, D_K$ is nonzero.*

Proof. $[x^N]P_K = \sum_r \big(\prod_j a_{i_j}\big)\ell^{\sum_j i_j}$ over representations r as in Lemma 3. The term of r has valuation $v_\mathfrak{p}(\ell)\sum_j i_j$. By Lemma 3 exactly one term has minimal valuation, so the sum is nonzero. □

Theorem 1. *For every $K \ge 1$ there is $f \in \mathbb{R}[x]$ of degree $D_K = 20(17^K - 1)/16$ with all $D_K + 1$ coefficients nonzero and $\|f^2\| \le 21 \cdot 9^{K-1}$. Hence $\hat{Q}(k) \le \frac{21}{9} k^{\log 9/\log 17}$ for $k = D_K + 1$.*

Proof. Take $f = P_K$ from Lemma 4. Scaling $x \mapsto \ell x$ does not change $\operatorname{supp}(p^2)$, so Lemma 2 applies. Since $k = D_K + 1 \ge 17^K$, we get $9^K \le k^{\log 9/\log 17}$. □

3 Remarks and limitations

- The same scheme with $n = 14$, $m = 12$, $S = [0, 4] \cup [12, 16] \cup [24, 28]$ gives $\log 7/\log 12 \approx 0.7831$ (also certified). More generally, the three-block family $n = 3r + 2$, $m = n - r/2$ gives $\log(r + 3)/\log m$, which is minimal at $r = 6$. Annealing over all symmetric patterns with $n \le 32$ and all bases found nothing better.

- The gain over Verdenius is asymptotic: our explicit bound $21 \cdot 9^{K-1}$ falls below $k^{\log 8/\log 13}$ only from $K = 7$ ($k \approx 5 \cdot 10^8$).
- **Rational coefficients:** none of the isolated real solutions we computed (all square symmetric systems with $n \leq 16$ and exponent below 0.816, and $n = 19, 20$) is rational up to rescaling of x, apart from Verdenius' own degree-8 seed. The bound over $\mathbb{Q}$ ($\log 6/\log 9$, relevant to Erdős Problem #485 as stated over $\mathbb{Q}$) is not improved.
- Extending the bound from this sequence to all k should follow from Erdős' interpolation argument [3]; we have not written this out.
- Heuristically, towers of a single seed whose zeros come from a square polynomial system seem limited to exponents around 0.77. Exponent $< 1/2$ (the Epoch AI "FrontierMath: Open Problems" target) would need a different source of cancellation.

4 Verification

Everything is reproducible from the repository: exhaustive and annealed pattern searches; carry-automaton growth rates (cross-checked by brute force); an independent tower check that does not import the search code (digit-sum counts $21 \cdot 9^{K-1}$ up to $K = 6$ by direct enumeration, and P_2 complete with $\|P_2^2\| = 189$ computed directly); and a brute-force check of Lemma 3 for $K \leq 4$.

5 Disclosure

The search, the certificates and this draft were produced with an AI assistant (Claude, Anthropic). The author has not yet had the arguments checked by an independent expert; comments and corrections are very welcome.

References

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